Physics of Complex Systems Homework 2

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(Dated: May 7, 2026)

Problem 1. Exercise 4.1: Convolution product **(6P)**

Let us consider the probability density of the gaps between Poisson-distributed events (the length x of silence between the clicks of a Geiger counter), which is defined by

$$p(x) = we^{-wx} \qquad (x \geq 0,\ w > 0).$$

(a) Compute the corresponding characteristic function (see lecture notes). (1P)

(b) Compute the two-fold convolution product $(p * p)(x)$ and the three-fold convolution product $(p * p * p)(x)$. (2P)

(c) Guess a general formula for the N-fold convolution product $p^{*N}(x)$ and prove it by recursion. (2P)

(d) Fit a normal distribution with mean μ and width σ to the probability density computed in (c) for large N.

Hint: Recall the behavior of cumulants. (1P)

Problem 2. Exercise 4.2: Harmonic oscillator **(6P)**

Consider a harmonic oscillator with eigenstates

$$|n\rangle = |0\rangle, |1\rangle, |2\rangle, \ldots$$

and eigenenergies

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right).$$

Let us assume that the eigenstates correspond to classical configurations and that the system evolves stochastically according to a continuous-time Markov process as follows:

$$n \to n+1 \qquad \text{with rate } \omega_\uparrow$$

$$n \to n-1 \qquad \text{with rate } \omega_\downarrow$$

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(a) Calculate the Liouville operator $\mathcal{L}$. (1P)

(b) Determine the left and the right eigenvector to the eigenvalue zero. (2P)

(c) Normalize the right eigenvector obtained in (b) so that it can be interpreted as a probability distribution. Under which condition is the normalization possible? (2P)

(d) In a canonical ensemble the probability of finding the system at energy E_n is proportional to

$$\exp\left(-\frac{E_n}{k_B T}\right),$$

where T is the temperature and k_B is the Boltzmann constant. Express the temperature T in terms of the rates $\omega_\uparrow$ and $\omega_\downarrow$. (1P)